PROBLEM 08

Kinematics, Two Trains on Tracks

Original statement · Problems.pdf, p. 8

Relative Motion

Two trains, A and B, are moving along parallel tracks. The separation between the tracks is a. The absolute velocity of each train is v_A and v_B, respectively. What is the relative velocity of each train with respect to the other?

Solve the same problem considering that the train B is now moving along a circular track of radius R. The minimum distance between the circular track and the straight track of A is a.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **Physical System:** Two trains A and B move on separate tracks with distance of closest separation a.
- Case 1: Two parallel straight tracks with constant speeds v_A and v_B.
- Case 2: Train A on a straight track, Train B on a circular track of radius R.
- **Parameters & Units:** Separation a $[\mathrm{m}]$, radius R $[\mathrm{m}]$, speeds v_A, v_B $[\mathrm{m/s}]$.
- **Hypotheses:** the tracks are in a plane; at $t = 0$ train A is at the origin and train B is at the point of its track closest to the track of A; the speeds are constant. The coordinates are the arc positions along the tracks (one per train). Both motions are prescribed, $x_A = v_A t$ and $\theta_B = v_B t/R$, so no degree of freedom is unknown (0 DOF): the problem is purely kinematic.
- **Observers.** In Case 1 the relative velocity is the same for any observer that translates with a train. In Case 2 an observer attached to B turns with it at $\omega_B = v_B/R$, so the velocity of A seen by this observer is derived in 3.4.

PHASE 2

Reference Frames & Vector Bases

- **Inertial Cartesian Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$.
- **Frame of train B (Case 2, rotating observer):** $S_B = \{B; \mathbf{i}_B, \mathbf{j}_B, \mathbf{k}\}$, turning with the train by the angle $\omega_B t$ about $\mathbf{k}$, $\omega_B = v_B/R$:

$$[{}_0R_B] = \begin{pmatrix} \cos\omega_B t & -\sin\omega_B t & 0 \\ \sin\omega_B t & \cos\omega_B t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_B] = \cos^2\omega_B t + \sin^2\omega_B t = 1$$

and $[{}_0R_B][{}_0R_B]^T$ has diagonal entries $\cos^2\omega_B t + \sin^2\omega_B t = 1$ (and 1) and off-diagonal entries $\cos\omega_B t \sin\omega_B t - \sin\omega_B t \cos\omega_B t = 0$, so it is I.

PHASE 3

Derivation

3.1 Kinematics

Case 1: Parallel Tracks

- Train A on line $y = 0$: $\mathbf{r}_A(t) = v_A t\,\mathbf{i}_0$.
- Train B on line $y = a$: $\mathbf{r}_B(t) = (x_{B0} + v_B t)\mathbf{i}_0 + a\mathbf{j}_0$.
- **Relative Velocity:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = (v_B - v_A)\mathbf{i}_0$$

$$\mathbf{v}_{A/B} = -\mathbf{v}_{B/A} = (v_A - v_B)\mathbf{i}_0$$

Case 2: Train B on a Circular Track of Radius R

- Straight track of A along the x-axis: $\mathbf{r}_A = v_A t\,\mathbf{i}_0$.
- Circular track center at $(0, R+a)$ so that the closest distance to A's track is $(R+a) - R = a$.
- Position of train B traveling at angular speed $\omega_B = \frac{v_B}{R}$ with angle $\theta_B(t) = \omega_B t$:

$$\mathbf{r}_B(t) = R\sin(\omega_B t)\mathbf{i}_0 + [R + a - R\cos(\omega_B t)]\mathbf{j}_0$$

- Velocity of train B:

$$\mathbf{v}_B = \frac{d\mathbf{r}_B}{dt} = R\omega_B\cos(\omega_B t)\mathbf{i}_0 + R\omega_B\sin(\omega_B t)\mathbf{j}_0 = v_B\cos\left(\frac{v_B t}{R}\right)\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Relative Velocity $\mathbf{v}_{B/A}$:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = \left[v_B\cos\left(\frac{v_B t}{R}\right) - v_A\right]\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Magnitude of Relative Velocity:**

$$v_{\text{rel}} = \sqrt{\left[v_B\cos\left(\frac{v_B t}{R}\right) - v_A\right]^2 + v_B^2\sin^2\left(\frac{v_B t}{R}\right)} = \sqrt{v_B^2 + v_A^2 - 2v_A v_B\cos\left(\frac{v_B t}{R}\right)}$$

3.2 Forces

Not applicable: kinematics-only problem (no masses or forces are given).

3.3 Equations of Motion

Not applicable: kinematics-only problem.

3.4 Integration and Answers

Answers. Case 1: $\mathbf{v}_{B/A} = (v_B - v_A)\,\mathbf{i}_0$ and $\mathbf{v}_{A/B} = (v_A - v_B)\,\mathbf{i}_0$. Case 2: $\mathbf{v}_{B/A} = [v_B\cos(v_B t/R) - v_A]\,\mathbf{i}_0 + v_B\sin(v_B t/R)\,\mathbf{j}_0$, with modulus $v_{\text{rel}} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B\cos(v_B t/R)}$. No integration is needed beyond the differentiation of 3.1.

Case 2, velocity of A with respect to an observer attached to B. The frame S_B of Phase 2 turns with $\boldsymbol{\omega}_{B0} = \omega_B \mathbf{k}$. Notes Eq. 3.26 gives $\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B - \boldsymbol{\omega}_{B0} \times (\mathbf{r}_A - \mathbf{r}_B)$. With $\mathbf{r}_A - \mathbf{r}_B = (v_A t - R\sin\omega_B t)\,\mathbf{i}_0 + (R\cos\omega_B t - R - a)\,\mathbf{j}_0$, $\mathbf{k}\times\mathbf{i}_0 = \mathbf{j}_0$, $\mathbf{k}\times\mathbf{j}_0 = -\mathbf{i}_0$ and $\omega_B R = v_B$, the three terms are

$$\mathbf{v}_A - \mathbf{v}_B = (v_A - v_B\cos\omega_B t)\,\mathbf{i}_0 - v_B\sin\omega_B t\,\mathbf{j}_0, \qquad -\boldsymbol{\omega}_{B0}\times(\mathbf{r}_A - \mathbf{r}_B) = -\omega_B\,(R + a - R\cos\omega_B t)\,\mathbf{i}_0 - \omega_B\,(v_A t - R\sin\omega_B t)\,\mathbf{j}_0$$

Adding them, $\omega_B R\cos\omega_B t = v_B\cos\omega_B t$ cancels the $-v_B\cos\omega_B t$ term and $\omega_B R\sin\omega_B t = v_B\sin\omega_B t$ cancels the $-v_B\sin\omega_B t$ term:

$$\mathbf{v}_{A/B} = \left[v_A - v_B\,\frac{R+a}{R}\right]\mathbf{i}_0 - \frac{v_A v_B t}{R}\,\mathbf{j}_0 \quad (\text{components in } \mathcal{B}_0)$$

It differs from $-\mathbf{v}_{B/A}$ because the observer attached to B rotates: at $t=0$, $v_{A/B} = v_A - v_B(1 + a/R)$, whereas the translating-observer value is $v_A - v_B$; for $R \to \infty$ both coincide.

PHASE 4

Interpretation & Checks

- **Dimensional check:** $[\mathbf{v}_{B/A}] = \mathrm{m/s}$; in Case 2 the argument of the trigonometric functions is $\dfrac{v_B t}{R} = \dfrac{(\mathrm{m/s})\,\mathrm{s}}{\mathrm{m}}$, which is dimensionless (rad), and $[v_B^2 + v_A^2 - 2v_A v_B\cos(\cdot)] = \mathrm{m^2/s^2}$, so v_{rel} is in m/s.
- **Role of the separation a:** it does not appear in any relative velocity, because $\mathbf{v}_{B/A}$ is the time derivative of $\mathbf{r}_B - \mathbf{r}_A$ and a only enters the constant offset of that vector.
- **Limiting cases of Case 2 (sympy):**
- $t = 0$: $v_{\mathrm{rel}} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B} = |v_B - v_A|$, because both velocities are parallel to $\mathbf{i}_0$ at that instant.
- $R \to \infty$ (the circular track straightens): $\cos(v_B t/R) \to 1$ and $v_{\mathrm{rel}} \to |v_B - v_A|$, which is the Case 1 result.
- $v_A = v_B = v$: $v_{\mathrm{rel}} = v\sqrt{2 - 2\cos(vt/R)} = 2v\left|\sin\dfrac{vt}{2R}\right|$, which starts at 0 and never exceeds $2v$.
- $v_A = 0$: $v_{\mathrm{rel}} = v_B$ for all t (relative to a stationary train, only the speed of B matters).

- **Symbolic check:** $\|\mathbf{v}_B - \mathbf{v}_A\|^2$ computed from the components equals $v_B^2 + v_A^2 - 2v_A v_B \cos(v_B t/R)$ with zero residual. The translating-observer results use only $\mathcal{B}_0$; the rotating observer of Case 2 uses $[{}_0R_B]$ (Phase 2). Dimensions of that result: $[v_A] = [v_B a/R] = [v_A v_B t/R] = \mathrm{m/s}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 8 in its solution keys, so the results are verified by independent derivation (the symbolic check of Phase 4).

Source: Problems.pdf